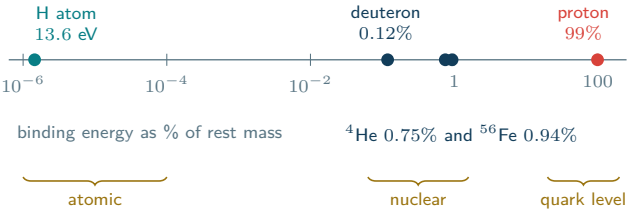
BINDING ENERGY ON ONE PAGE

Mass defect and binding energy

Why a bound system weighs less than its parts, from 13.6 eV in hydrogen to 28.3 MeV in helium-4.

1 / THE RULE AND THE PICTURE

The energy released on binding is the missing mass times c^2



Atoms and nuclei weigh less than their separate parts, and the difference is the energy released when they formed:

$$B = \Delta m\, c^2, \quad \Delta m = \sum m_{\text{parts}} - m_{\text{bound}}$$

Inside a proton the sign reverses, since the three quark masses sum to only 8.8 MeV of its 938.3 MeV, so 99% of the proton's mass is strong-interaction energy.

2 / ATOMIC SCALE / HYDROGEN

$m_H c^2 = (m_p c^2 + m_e c^2) - 13.6$ eV, and a photon is emitted

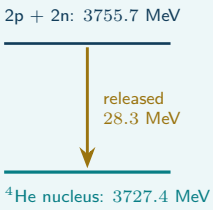
$m_p c^2 + m_e c^2$	$938.272 + 0.511 = 938.783$ MeV
mass defect 13.6 eV/c^2	2.42×10^{-35} kg
fraction of the mass	1.4×10^{-8}
photon emitted, $\lambda = hc/E$	91.2 nm (ultraviolet)

Bringing a proton and electron of opposite charge together does negative work, so energy leaves as a photon and the atom weighs less than its parts: $m_{\text{atom}} = m_{\text{nucleus}} + Zm_e - B_e/c^2$, with B_e counted positive.

The defect is far too small to weigh, which is why atomic binding is read from spectra and nuclear binding from masses: all 79 eV of helium's electrons is 2×10^{-6} of the 28.3 MeV of its nucleus.

3 / NUCLEAR SCALE / THE FIVE-STEP PROCEDURE

Atomic mass, then nuclear mass, then mass defect, then MeV, then per nucleon

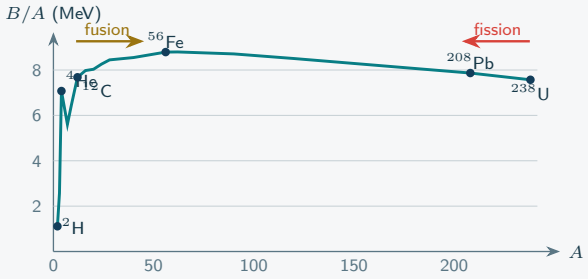


	^{4}He	^{63}Cu
m_{atom} (u)	4.002602	62.929597
$m_{\text{nucleus}} = m_{\text{atom}} - Zm_e$	4.001504	62.913676
$\Delta m = Zm_p + Nm_n - m_{\text{nucleus}}$	0.030378	0.591938
$B = \Delta m \times 931.5$ MeV/u	28.30 MeV	551.4 MeV
B/A	7.07 MeV	8.75 MeV

With 1 u = 931.5 MeV/c^2 no kilograms are needed, and using hydrogen atoms, $\Delta m = Zm_H + Nm_n - m_{\text{atom}}$, cancels the electrons and gives the same 0.030378 u for helium.

4 / THE CURVE / WHICH REACTIONS RELEASE ENERGY

Binding energy per nucleon peaks near iron and nickel



Light nuclei gain binding per nucleon when they fuse, and heavy ones gain it when they split, so both reactions move towards the peak and release energy. Four hydrogen atoms fusing to one helium atom lose 0.028697 u = 26.7 MeV, which is 0.71% of their mass, and iron sits at the top with nothing left to release. Total energy released is $\sum B_{\text{products}} - \sum B_{\text{reactants}}$, using total B and not B/A.

5 / DECAYS / ENERGY RELEASED AS KINETIC ENERGY

Alpha decay from binding energies, and positron decay from atomic masses

decay	working	energy	alpha share
^{223}Ra $\to$ ^{219}Rn $+^4$He	$(219\times7.7238 + 4\times7.074) - 223\times7.6853$	6.0 MeV	5.9 MeV
^{238}U $\to$ ^{234}Th $+^4$He	$(234\times7.6042 + 4\times7.074) - 238\times7.5901$	4.27 MeV	4.20 MeV
^{18}F $\to$ ^{18}O $+e^+ + \nu_e$	$\Delta m\, c^2 - 2m_e c^2 = 1.656 - 1.022$	0.634 MeV	shared with ν_e

The parent is at rest, so momentum conservation gives the alpha the fraction $M_{\text{daughter}}/(M_{\text{daughter}} + M_\alpha)$ of the energy; less energy means a longer half-life, 11.4 days for radium-223 against 4.5×10^9 years for uranium-238.

ASIDE / HEAVIER BY BINDING

In the quark table the proton, neutron and helium are 99.06%, 98.77% and 98.91% strong-interaction energy.

ASIDE / CHECK IN PYTHON

`(2*1.007276+2*1.008665-(4.002602-2*0.000549))*931.5` gives 28.30.

Convention. Binding energies are positive; 1 u = 931.5 MeV/c^2. After the slides E29 to E39c and the binding-energy learner.